Learned Geometry for Time-Series Anomaly Detection

DG_AD: where we are (Week 15)

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Our method: a denoiser and a metric network

window y (length L)

↓

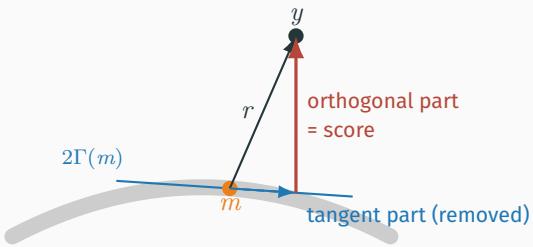
denoiser $m(y)$
nearest healthy point

↓

metric network $2\Gamma(m)$
tangent projector at m

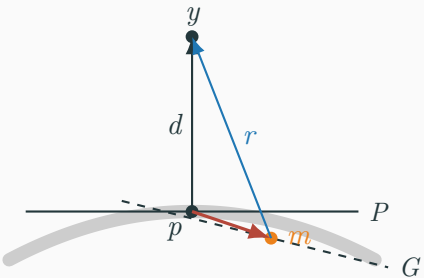
↓

score $s = \|(I - 2\Gamma(m))\,(y - m)\|$



Rare is not abnormal: only the distance *off* the set counts.

Bound on Anomaly Score



denoiser error

y window; p its nearest healthy point
m denoised point $m(y)$; $r = y - m$
d true distance $\|y - p\|$; s our score
P true tangent projector at p
$N = I - P$, true orthogonal projector at p
$G = 2\Gamma(m)$, learned tangent projector

1. Score and distance:

$$s = \|(I - G)\, r\|, \qquad d = \|y - p\| = \|N(y - p)\|$$

($y - p$ is orthogonal to the set, so N leaves it unchanged)

2. Use $r = (y - p) - (m - p)$ and $I - G = N + (P - G)$:

$$(I - G)\, r = \underbrace{N(y - p)}_{\text{length } d} - \underbrace{N(m - p)}_{\text{denoiser error}} + \underbrace{(P - G)\, r}_{\text{projector error}}$$

3. The first term has length d, so s differs from d by at most the lengths of the other two:

$$\boxed{|s - d| \;\le\; \|N(m - p)\| + \|G - P\|\,\|r\|}$$

Toy shapes, exact quantities: bound holds, close to tight (median 0.88).

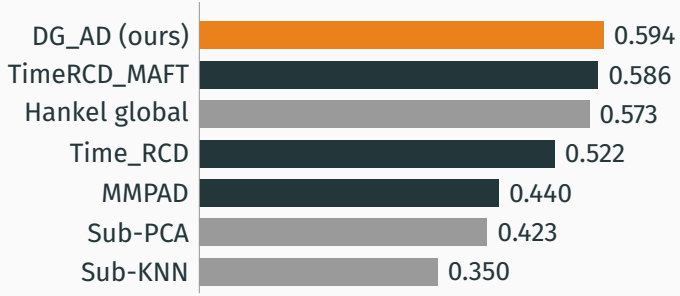
Held-out result: level with the leader

0.594

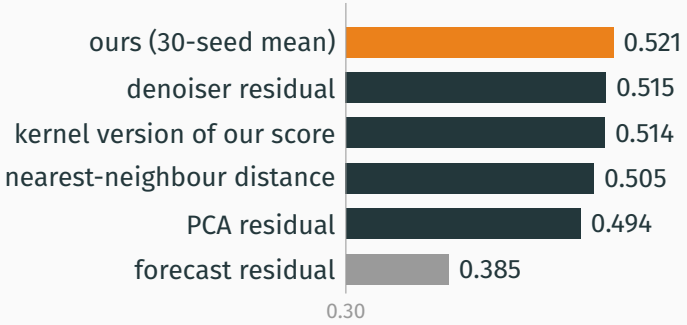
mean VUS-PR, Eva (350 series)

vs. leader:

$+0.008\ [-0.031, +0.046]$

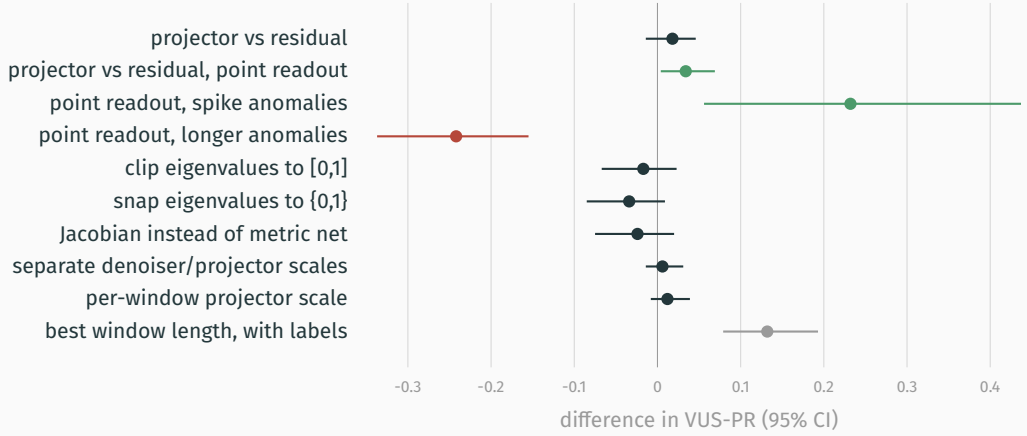


Honest context: simple distances, same pipeline

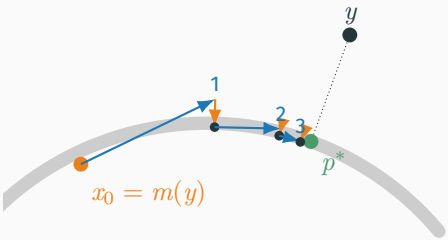


Tuning-48. Biggest lever: window $\rightarrow$ time-point mapping ($+0.05$ to $+0.07$).

Ablation: what moves the score (Tuning-48)



Next: make the denoiser project



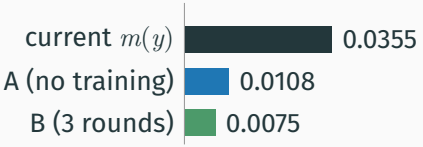
→ tangent step $x \leftarrow x + 2\Gamma(x)(y - x)$

→ pull-back $x \leftarrow m(x)$

schematic, half-size steps

A alternate the two steps: Riemannian gradient descent on $\frac{1}{2}\|x - y\|^2$ with a learned retraction

B train m to output A's answer directly



Noisy sine: B diverges (diagnosis running).

Idea: projection as a drift

1. Functional (per query y; $\hat{G} = 2\Gamma$ pretrained):

$$F_y(x) = \tfrac{1}{2}\|\hat{G}(x-y)\|^2 + \tfrac{\lambda}{2}\,\mathrm{dist}(x,\mathcal{M})^2$$

2. Drift $V = -\nabla F_y = V^+ - V^-$:

$V^+(x) = \lambda\,(I-\hat{G})\,(m_0(x)-x)$ onto the set $\qquad V^-(x) = \hat{G}\,(x-y)$ offset from y

3. JKO, transport cost in the tangent plane:

$x_{n+1} = \arg\min_x F_y(x) + \frac{1}{2\tau}\|\hat{G}(x-x_n)\|^2 \;\Rightarrow$ tangent step + projection $\to\ p^*$

4. Train (frozen target): $L(\theta) = \mathbb{E}\,\|m_\theta(y) - \mathrm{sg}[x+\tau\,V(x)]\|^2, \quad x = m_\theta(y)$